

Mean Values

Given a data set $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, $\mathbf{x}_n \in \mathbb{R}^D$, we compute the mean of the data set as

$$\mathbb{E}[\mathcal{D}] = \frac{1}{N} \sum_{n=1}^N \mathbf{x}_n$$

Variances of 1D data sets

Given a data set $\mathcal{D} = \{x_1, \dots, x_N\}$, $x_n \in \mathbb{R}$, we compute the variance of the data set as

$$\mathbb{V}[\mathcal{D}] = \frac{1}{N} \sum_{n=1}^N (x_n - \mu)^2$$

where μ is the mean value of the data set.

Variances of higher-dimensional data sets

Given a data set $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, $\mathbf{x}_n \in \mathbb{R}^D$, we compute the variance of the data set as

$$\mathbb{V}[\mathcal{D}] = \frac{1}{N} \sum_{n=1}^N (\mathbf{x}_n - \boldsymbol{\mu})(\mathbf{x}_n - \boldsymbol{\mu})^\top \in \mathbb{R}^{D \times D}$$

where $\boldsymbol{\mu} \in \mathbb{R}^D$ is the mean value of the data set.

Effect of Linear Transformations

Consider a data set $\mathcal{D} = \{\mathbf{x}_1, \dots, \mathbf{x}_N\}$, $\mathbf{x}_n \in \mathbb{R}^D$, with

$$\mathbb{E}[\mathcal{D}] = \boldsymbol{\mu}$$

$$\mathbb{V}[\mathcal{D}] = \boldsymbol{\Sigma}$$

If we now modify every $\mathbf{x}_i \in \mathcal{D}$ according to

$$\mathbf{x}'_i = \mathbf{A}\mathbf{x}_i + \mathbf{b}$$

for a given $\mathbf{A}, \mathbf{b}$, then

$$\mathbb{E}[\mathcal{D}'] = \mathbf{A}\boldsymbol{\mu} + \mathbf{b}$$

$$\mathbb{V}[\mathcal{D}'] = \mathbf{A}\mathbf{Q}\mathbf{A}^\top$$

where $\mathcal{D}' = \{\mathbf{x}'_1, \dots, \mathbf{x}'_N\}$

Dot product

The **dot product** is defined as

$$\mathbf{x}^\top \mathbf{y} = \sum_{d=1}^D x_d y_d, \quad \mathbf{x}, \mathbf{y} \in \mathbb{R}^D.$$

- ▶ The **length** of $\mathbf{x}$ is then

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}}.$$

- ▶ The **angle** ω between two vectors $\mathbf{x}, \mathbf{y}$ can be computed using

$$\cos \omega = \frac{\mathbf{x}^\top \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

Inner product

Consider a vector space V . A positive definite, symmetric bilinear mapping $\langle \cdot, \cdot \rangle : V \times V \rightarrow \mathbb{R}$ is called an **inner product** on V .

- ▶ **symmetric**: For all $x, y \in V$ it holds that $\langle x, y \rangle = \langle y, x \rangle$
- ▶ **positive definite**: For all $x \in V \setminus \{0\}$ it holds that $\langle x, x \rangle > 0$, $\langle 0, 0 \rangle = 0$
- ▶ **bilinear**: For all $x, y, z \in V, \lambda \in \mathbb{R}$

$$\langle \lambda x + y, z \rangle = \lambda \langle x, z \rangle + \langle y, z \rangle$$

$$\langle x, \lambda y + z \rangle = \lambda \langle x, y \rangle + \langle x, z \rangle$$

Inner product: Lengths and distances

Consider a vector space V with an inner product $\langle \cdot, \cdot \rangle$.

- ▶ The **length** of a vector $x \in V$ is

$$\|x\| = \sqrt{\langle x, x \rangle}$$

- ▶ The **distance** between two vectors $x, y \in V$ is given by

$$d(x, y) = \|x - y\| = \sqrt{\langle x - y, x - y \rangle}$$

Inner product: Angles

Consider a vector space V with an inner product $\langle \cdot, \cdot \rangle$.

The **angle** ω between two vectors $x, y \in V$ can be computed via

$$\cos \omega = \frac{\langle x, y \rangle}{\|x\| \|y\|}$$

where the length/norm

$$\|x\| = \sqrt{\langle x, x \rangle}$$

is defined via the inner product.

Projection onto 1D subspaces

Consider a vector space V with the dot product at the inner product and a subspace U of V . With a basis vector $\mathbf{b}$ of U , we obtain the **orthogonal projection** of any vector $\mathbf{x} \in V$ onto U via

$$\pi_U(\mathbf{x}) = \lambda \mathbf{b}, \quad \lambda = \frac{\mathbf{b}^\top \mathbf{x}}{\mathbf{b}^\top \mathbf{b}} = \frac{\mathbf{b}^\top \mathbf{x}}{\|\mathbf{b}\|^2}$$

where λ is the **coordinate** of $\pi_U(\mathbf{x})$ with respect to $\mathbf{b}$.

The **projection matrix** $\mathbf{P}$ is

$$\mathbf{P} = \frac{\mathbf{b}\mathbf{b}^\top}{\mathbf{b}^\top \mathbf{b}} = \frac{\mathbf{b}\mathbf{b}^\top}{\|\mathbf{b}\|^2}$$

such that

$$\pi_U(\mathbf{x}) = \mathbf{P}\mathbf{x}$$

for all $\mathbf{x} \in V$.

Projection onto k -dimensional subspaces

Consider an n -dimensional vector space V with the dot product at the inner product and a subspace U of V . With basis vectors $\mathbf{b}_1, \dots, \mathbf{b}_k$ of U , we obtain the **orthogonal projection** of any vector $\mathbf{x} \in V$ onto U via

$$\pi_U(\mathbf{x}) = \mathbf{B}\boldsymbol{\lambda}, \quad \boldsymbol{\lambda} = (\mathbf{B}^\top \mathbf{B})^{-1} \mathbf{B}^\top \mathbf{x}$$

$$\mathbf{B} = (\mathbf{b}_1 | \dots | \mathbf{b}_k) \in \mathbb{R}^{n \times k}$$

where $\boldsymbol{\lambda}$ is the **coordinate vector** of $\pi_U(\mathbf{x})$ with respect to the basis $\mathbf{b}_1, \dots, \mathbf{b}_k$ of U .

The **projection matrix** $\mathbf{P}$ is

$$\mathbf{P} = \mathbf{B}(\mathbf{B}^\top \mathbf{B})^{-1} \mathbf{B}^\top$$

such that

$$\pi_U(\mathbf{x}) = \mathbf{P}\mathbf{x}$$

for all $\mathbf{x} \in V$.

Key steps of PCA algorithm

1. Compute the mean μ of the data matrix $\mathbf{X} = [\mathbf{x}_1 | \dots | \mathbf{x}_N]^\top \in \mathbb{R}^{N \times D}$
2. Mean subtraction: Replace all data points $\mathbf{x}_i$ with $\tilde{\mathbf{x}}_i = \mathbf{x}_i - \mu$.
3. Divide the data by its standard deviation in each dimension:
 $\bar{\mathbf{X}}^{(d)} = \tilde{\mathbf{X}} / \sigma(\mathbf{X}^{(d)})$ for $d = 1, \dots, D$.
4. Compute the eigenvectors (orthonormal) and eigenvalues of the data covariance matrix $\mathbf{S} = \frac{1}{N} \bar{\mathbf{X}}^\top \bar{\mathbf{X}}$
5. Choose the eigenvectors associated with the M largest eigenvalues to be the basis of the principal subspace.
6. Collect these eigenvectors in a matrix $\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_M]$
7. Orthogonal projection of the data onto the principal axis using the projection matrix $\mathbf{B}\mathbf{B}^\top$

PCA in high dimensions

- ▶ We need to solve the eigenvector/eigenvalue equation

$$\frac{1}{N} \bar{\mathbf{X}} \bar{\mathbf{X}}^\top \mathbf{c}_i = \lambda_i \mathbf{c}_i$$

where $\mathbf{c}_i = \bar{\mathbf{X}} \mathbf{b}_i$

- ▶ We want to recover the original eigenvectors $\mathbf{b}_i$ of the data covariance matrix $\mathbf{S} = \frac{1}{N} \bar{\mathbf{X}}^\top \bar{\mathbf{X}}$
- ▶ Left-multiplying eigenvector equation by $\bar{\mathbf{X}}^\top$ yields

$$\underbrace{\frac{1}{N} \bar{\mathbf{X}}^\top \bar{\mathbf{X}}}_{=\mathbf{S}} \bar{\mathbf{X}}^\top \mathbf{c}_i = \lambda_i \bar{\mathbf{X}}^\top \mathbf{c}_i$$

and we recover $\bar{\mathbf{X}}^\top \mathbf{c}_i$ as an eigenvector of $\mathbf{S}$ with (the same) eigenvalue λ_i

Note: To perform PCA as discussed in the lecture we need to make sure that $\|\bar{\mathbf{X}}^\top \mathbf{c}_i\| = 1$.